

ECON100B - SECTION 11

The Labor Market

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 2 due tomorrow, Friday, 2/27
- In-Class Midterm, 3/5

Agenda

- The Labor Market

Question of the Day

- (e) [2 points] Recall that when production is Cobb-Douglas with a capital share $\alpha = 1/3$, the marginal product of capital is given by:

$$MPK = \frac{1}{3} \bar{A} \left(\frac{L}{K} \right)^{2/3} \quad (1.2)$$

Suppose we knew these calibrated¹ values for A, K, and L:

A = 1,983

K = 80,823,096

L = 163

“Total Factor Productivity” is a residual

The U.S. capital stock is about \$80 trillion

U.S. employment is about 163 million

Using a calculator or a spreadsheet, insert these values into the right-hand side of equation (1.2) and report and comment on what you calculated. In words, what does your value mean? Round to the nearest thousandth = tenth of a percent and interpret.

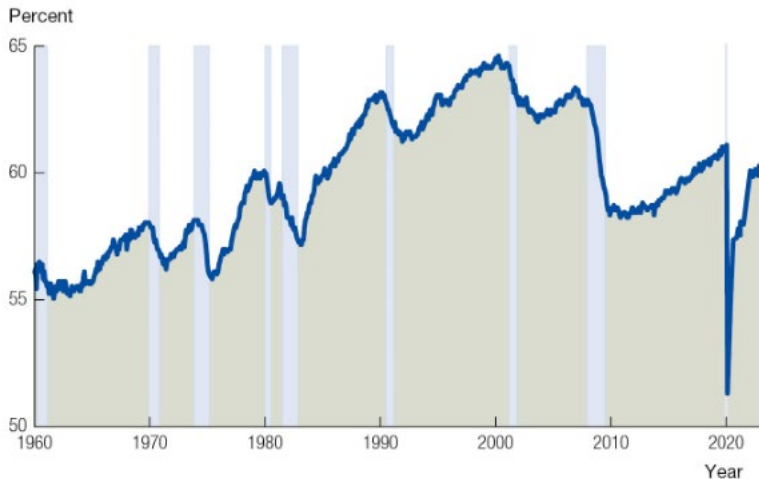
What is something good or interesting that you’ve read recently?

The Labor Market



The Labor Market, Introduction

Ratio of Employment to Population in the United States



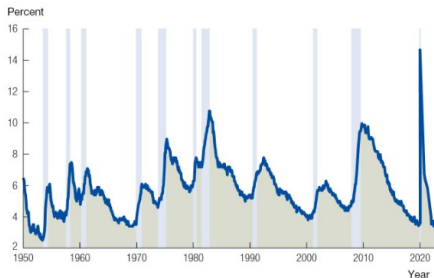
Review on Unemployment

A person is considered to be unemployed if they:

- Don't have a job that pays a wage or salary
- Have actively looked for a job in the past 4 weeks
- Currently available to work

The Labor Force is equal to the count of Employed + Unemployed

- Unemployment rate is calculated as $\text{number of unemployed} / \text{size of labor force}$



Review on Unemployment

Some things to consider on employment:

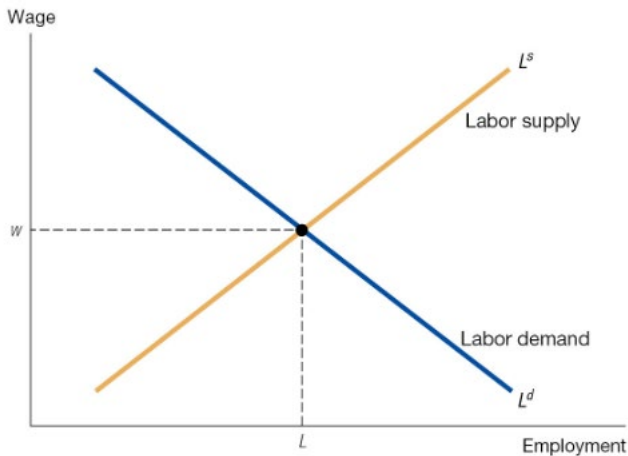
- Jobs are being created and destroyed all of the time: this is what's known as "gross flows" and is a normal part of the modern economy
- Most unemployment is relatively short, lasting under 14 weeks, but for some people, unemployment can last a very long time
- Balancing incentives to work and social safety is a big challenge in unemployment/labor policy

Several types of unemployment:

- Frictional unemployment: natural unemployment occurring from changing jobs in a dynamic economy
- Structural unemployment: results from features of the labor market, such as hiring and firing costs, minimum wage, and unemployment benefits
- Cyclical unemployment: not natural unemployment, from economic shocks

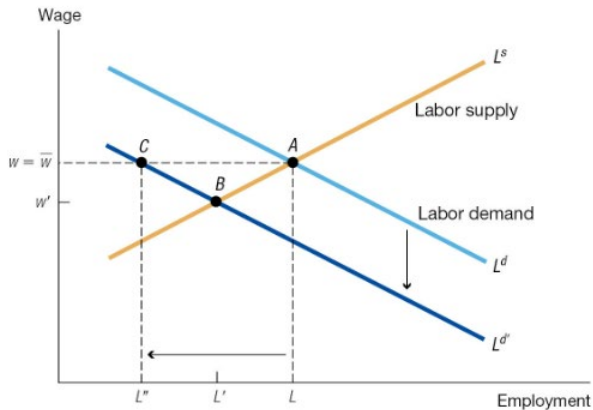
The Labor Market

The Labor Market



Wage Rigidity

A Reduction in Labor Demand with Wage Rigidity



Efficiency Wages

- Refer to the idea that paying your workers more than the bare minimum might be a good idea for output (who would've thought)
- Workers are better fed and have more of their basic needs met
- Workers may feel as if their work is more valuable and thus put in more effort
- Difficult question is where to exactly put wages as a “profit-maximizing” firm

Labor Market Practice

1. Say I have been out of a job for 2 months now. I spent the first 4 weeks looking for a job but haven't been looking since then. How does my decision to not look for a job anymore affect the unemployment rate?
2. Why is the unemployment rate potentially a poor metric to use? What are some other metrics that might better capture the strength of an economy's labor market?
3. Using the Supply & Demand Labor Market model, what would happen if there was a shock in increased oil prices? Do any curves shift and if there a new equilibrium in the model?
4. Using the wage rigidities model, how might wage rigidities explain the massive spike in unemployment during the Great Depression?

Practice Answers

1. I would no longer be counted as unemployed because I have not been looking for a job
2. It doesn't count people who might be interested in working but have not been searching recently. A better metric might be Labor Force Participation Rate/Ratio of Employment to Population, or other metrics
3. Demand shifts left, correspondingly resulting in lower wages and employment
4. It could help explain because there is more than a natural drop in unemployment, as there is no pressure valve release of the wages dropping

The Bathtub Model

$$E_t + U_t = \bar{L}, \quad (7.2)$$

$$\Delta U_{t+1} = \underbrace{\bar{s}E_t}_{\substack{\text{employed people} \\ \text{who lose their jobs}}} - \underbrace{\bar{f}U_t}_{\substack{\text{unemployed people} \\ \text{who find new jobs}}} \quad (7.3)$$

The Bathtub Model Steady State

$$\begin{aligned} 0 &= \bar{s}E_t - \bar{f}U_t \\ &= \bar{s}(\bar{L} - U_t) - \bar{f}U_t \\ &= \bar{s}\bar{L} - (\bar{f} + \bar{s})U_t. \end{aligned}$$

$$U^* = \frac{\bar{s}\bar{L}}{\bar{f} + \bar{s}}.$$

$$u^* \equiv \frac{U^*}{\bar{L}} = \frac{\bar{s}}{\bar{f} + \bar{s}}.$$

Solving the bathtub model's steady state gives us the “natural rate” of unemployment

Global labor markets

Global labor markets

- Most places around the world including Europe have higher unemployment rates than in the United States
 - European countries often have better unemployment and welfare benefits
 - According to (some) economists, these are “inefficient” labor markets by providing incentives
- Some places, like Japan, have lower rates
- Europeans tend to work 20-25% hours less than Americans while East Asian countries tend to work 10-20% more hours

Present Discounted Value

- Present discounted value is the value of a future amount in current terms
- Under this example of 5 years and 1,000 dollars, we plug in R as a discount rate (often the interest rate)

$$pdv = \frac{\$1,000}{(1 + R)^5}.$$

- You can also think about it as the amount you have to deposit today to get \$1,000 in 5 years
- How about instead of working backwards, we say we're given a certain amount every year?

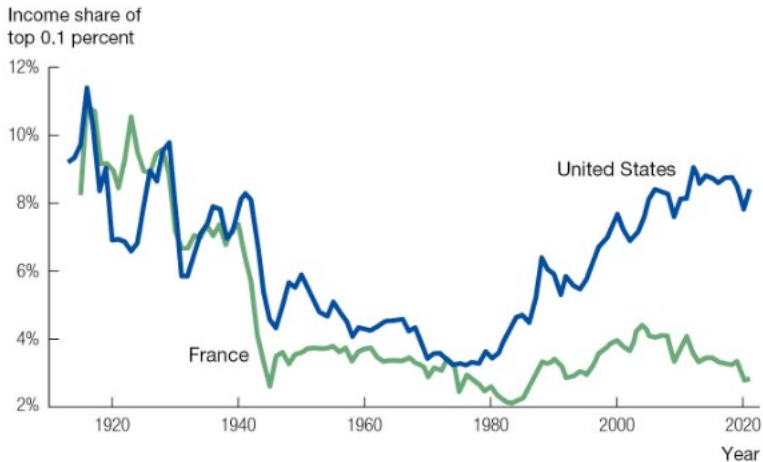
$$pdv = \$100 \times \frac{1 - \left[\frac{1}{(1 + R)} \right]^{20}}{1 - \frac{1}{(1 + R)}}.$$

PDV in action

$$pdv = \$100,000 \times \frac{1 - \left(\frac{1}{1+R}\right)^{45}}{1 - \frac{1}{1+R}}$$
$$= \$2.53 \text{ million.}$$

- Using PDV, we can justify the returns on attending college
- College-educated workers earn on average 50% higher than their non college-educated counterparts
- The grow in the college/non-college wage gap can potentially be explained by skill-biased technical change and globalization

Income inequality in the U.S. vs France



Labor Market Practice

1. Using the bathtub model, what would be the steady state unemployment rate in an economy if we assumed that the rate “s” of job separation was 1% per month and the rate of finding a job “f” was 20% per month?

$$u^* \equiv \frac{U^*}{\bar{L}} = \frac{\bar{s}}{\bar{f} + \bar{s}}.$$

2. Say I've been awarded a \$10,000 grant, but one that I can only receive after 4 years of study. What is the present discounted value of this grant given a 5% discount/interest rate?
3. What are economist's theories for why the college/non-college wage gap is growing? Do you think this gap will continue to grow, and what consequences will that have?

Practice 2 Answers

1. Roughly 5%
2. 8227
3. The most common theories are globalization and the growth of skill-biased jobs. Up to you on what you think future consequences are, but I would say something has to change, otherwise it will become a very politically and economically untenable situation!